Rupali Bhatnagar

Senior Software Engineer

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Work Experience

ServiceNow	Sr. Software Engineer (Security Operations)	Mar'22 – May'25

Individual contributor on Security Operations and Vulnerability Response, building on the ServiceNow© platform. Underlined features link to ServiceNow's public documentation.

- **Exception Management — Approval Rule Engine** for vulnerability exceptions — designed a multi-level, dynamic approval rule engine on ServiceNow Flow Designer supporting deferment and exception requests across differing enterprise policies; Flow Designer approvals shipped in **v15.0.2**. **Reduced approval time by 40%.**
- **Penetration Testing Request Catalog & Workflow Automation** — built the request catalog and automated intake, triage and routing for penetration-test requests across **50+ subsystems**, **cutting manual coordination by more than 30%** — the testing itself stays expert and manual by design.
- **Background Job Processing Framework** — architected the multithreaded framework for long-running tasks that previously strained system resources, partitioning work into parallel chunks, with migration via a configuration interface. Rollup jobs were converted to multithreaded background jobs in **v18.2.5**:
  - Multithreaded execution — **60% efficacy improvement** in some use cases, **1M+ tasks/month**.
  - Job status tracking and cancellation — **100% visibility** on executing and errored tasks.
  - Configurable **retry logic and dead-letter queuing** for failed jobs.
- **Real-Time Data Aggregation Engine** — high-performance aggregation framework for near real-time multi-dimensional analytics, powering key internal dashboards over **100K+ records in an 8-hour window**.

ServiceNow	Software Engineer (Security Operations)	Apr'19 – Mar'22

- **Assignment Rule Engine for Vulnerability Remediation** — cache-optimised engine prioritising infrastructure vulnerabilities and policy violations and routing **vulnerable items (VITs)** to the correct remediation groups via the Glide platform. Used daily by ITIL users; handles **1M+ vulnerable items daily**.
- **Assignment Recommendations on Predictive Intelligence** — classification models on the **Predictive Intelligence** framework with feature engineering and a reviewer-facing recommendation interface across **vulnerable items and vulnerability groups**; shipped in **v12.1.4**. **20% increase in task allocation accuracy.**
- **CISO & Vulnerability Dashboards** — led design and development of the CISO dashboard and supporting vulnerability dashboards, defining the key metrics and structuring the data pipeline. Used on a regular basis by **50+ organisations**.
- **Vulnerability Source Integration** — engineered integrations with **Veracode, Qualys, Tenable and Microsoft TVM**, building the **ETL pipelines** that standardise **500K+ vulnerability records/day** into a common model, with error handling and retry for reliable ingestion into downstream remediation workflows.
- **POA&M Generator** — auto-generated **Plans of Action & Milestones**, the remediation-plan artifact required under **NIST SP 800-53** and **FedRAMP**, directly from vulnerability findings rather than from control failures. Built at an internal hackathon and **adopted by the Now on Now support team**, who ran it against ServiceNow's own instance.
- **Hackathon Projects**:
  - **Diagnostic Framework** — internal support tooling that streamlined issue tracking and reproduction across customer instances, **reducing debugging time by 40%**.
  - **ML over Vulnerability Response data** — Predictive Intelligence **clustering** to group vulnerable items beyond the product's filter-based group rules, and **regression** to predict time-to-remediate, where shipped remediation targets are deterministic date arithmetic.
  - **High-Volume Data Generator** — scalable generation of **100M+ records across 20–30 tables** for load testing; used by the performance team.

Darwinbox Digital Solutions	Software Developer	Aug'18 – Mar'19

- Designed and implemented a microservice architecture capturing an **audit trail** of client data modifications, written in **PHP with the Yii framework**, listening to **MongoDB oplogs** and publishing to an **AWS SQS** queue.
- Built an **Oplog Processor in Node.js** diffing strings, arrays and embedded documents into human-readable audit entries, with an internal dashboard for adding trackable fields.

NowFloats	Software Developer	Jun'17 – Jul'18

- Primary work was the **Boost merchant portal's customer-relationship surface** — customer enquiries, call logs and call tracking — built in **C#/ASP.NET, Bootstrap, JavaScript, jQuery and Knockout.js** and serving **50,000+ merchant websites**.
- Integrated the **Facebook Graph API** so merchants could publish products and updates to their Facebook profile, page and shop (Ads extension).
- Built the **SPIDER** module using **machine learning** to predict sales revenue per representative from historical performance, with up to **84% accuracy**.
- Built an **offers and discounted-purchase flow** on **Kitsune**, NowFloats' serverless application framework, wiring in **k-Pay** and **Assured Purchase** — the escrow-backed payment-protection system that held buyer funds until an order was confirmed fulfilled — and surfaced it to merchants through the **Boost Keyboard**; also built the **Order Management** dashboard (**AngularJS, C#**) covering purchase, payment and delivery status and grievance redressal.
- Worked on the **appointment-scheduling system** across the Kitsune-built merchant websites — per-staff availability and slot modelling, booking and management — in **Node.js and C#**, generic across the clinic, salon, restaurant and hotel verticals, where it ships today as an appointment scheduler for practices and a room-reservation engine for hotels.

Finomena (wound down 2017)	Software Developer	Feb'17 – May'17

- Built the internal dashboards for **Loan Management**, mandate (auto-debit) management, collections and transaction tracking on the small-ticket lending platform, in **PHP (Laravel)**, scaling to **8,000+ loan applications/month** with Eloquent ORM over MySQL.
- Built the **message categorisation service** feeding Finomena's alternative underwriting — the app read the borrower's SMS inbox and the service parsed and classified message content into structured credit signal, for a lender whose model scored borrowers with **no credit history**.
- Built the internal dashboard for the city **runner** fleet — the field agents handling collections and repayment follow-up.
- Used **Artisan** for task scheduling, cron handling and testing; Laravel modules for authorisation, validation, event handling and mailing.

Babajob.com (acquired by Quikr, Jun'17)	Software Engineer	May'16 – Jan'17

- Built **TelephonySystem**, the **Golang** service behind the missed-call job channel and consumed by Babajob's C# platform — inbound and outbound call handlers against **Exotel** numbers, call-rotation strategy, incoming-call handling and the logging subsystem. **Stateful by design**: a first call created the jobseeker's profile, and later calls were recognised by source number and enriched it — so a feature-phone user built a profile by ringing a number. **10,000+ inbound calls and 20,000+ outbound SMS per day** on **RabbitMQ**, **Redis** and **MongoDB**; one of two developers on a five-person telephony team.
- Wired that profile into an **automated matching and application pipeline**: each completed call published the jobseeker profile to the outbound messaging service, which matched it against open roles through **Elasticsearch**, sent the shortlist by SMS, then walked the seeker through a sequential application that connected them to the job creator.
- Contributed to the **search subsystem in Java over Elasticsearch**, with dynamic query generation via Mustache templates.

Selected Projects — 2025 to present

Independent and consulting engineering through **Tathya Dev** (tathya.dev) — **10+ production web products across 43 repositories and 4,238 commits in 12 months**, enabled by AI-augmented engineering workflows.

AI & agent tooling	
propagate Claude Code plugin · Node, ESM	• Watches **~920 declared file couplings across 14 workspaces**, deriving drift from file *content* on demand rather than storing state — github.com/Rupali59/propagate. Hooks run in every agent session on the machine, delivering the matching known hazard before a risky command executes. • Never edits a downstream file — it finds and explains, a person approves, the decision lands in an append-only ledger. Every detector ships a `selftest` asserting it can fire; auditing its own monitor to that standard surfaced **49 crashes in a stderr log nothing read**. **142 tests.**
LLM Classification Pipeline Demand-intelligence ingestion · Gemini · schema-constrained	• Built a production LLM classification pipeline: Gemini under enforced response schemas at `temperature: 0`, batched, determinism from cache-on-first-write rather than the model, and a keyword-classifier fallback so a provider outage degrades quality instead of breaking ingestion. • Model-proposed sources are **existence-checked against the live API before acceptance**; an inconclusive check flags rather than asserts.
Products & platforms	
Multi-tenant CRM & Workflow Platform Service-business CRM · Express · Next.js 14 · BullMQ	• Built an Express API and two Next.js 14 applications with a **database-driven RBAC/ACL engine**: runtime-editable roles, generic field-level access control over CRUD, per-actor tier projection, fail-closed reconciliation and a capability-matrix UI that shows the consequences of a change before an atomic save. • WebAuthn + OTP, BullMQ/Redis queue, workflow engine, web-push. Six-phase engine cutover that deleted two legacy sweep passes only after verifying their outcomes equal to the new engine's and freezing them as a golden test — the cutover surfaced **five defects that existed only because the compiled graph had never executed**. • Suite **104s → 28s**. **228 test files.**

Content & SEO Platform Practitioner marketing site · Next.js 16 · TypeScript	• Shipped a production Next.js site on the computation service below. • Nine templated landing pages plus a hub on a shared shell, `FAQPage` structured data with selective `noindex`, a server-side analytics pipeline, signed-cookie location handling with a TTL-indexed response cache, and public pages migrated to static generation.
Practitioner Admin Console Scheduling, CRM and CMS · Next.js 16 · MongoDB	• Built the practitioner-facing console behind the public site — **~30 routes** over **26 Mongoose models**: schedule in day, week, month, list and waitlist views, a clients CRM with per-client kundli and birth-detail revert, tasks, notes, an articles CMS and fourteen settings and site-content editors. • **WebAuthn passkeys** with Twilio OTP fallback and JWT sessions; practitioner notes **encrypted at rest with AES-256-GCM** under a documented dual-read key-rotation procedure.
Demand-Intelligence Platform Multi-source ingestion & scoring · Next.js · Mongoose · Gemini	• Built multi-source ingestion and opportunity scoring over Reddit, Google Trends, YouTube, Search Console and GA4, fed by the Gemini pipeline above, with an entity ontology mapping demand against supply. • Search Console authenticated via **Workload Identity Federation** to remove service-account key fragility.
Ephemeris Computation Service Swiss Ephemeris over FastAPI · Python 3.12 · MongoDB	• Engineered a FastAPI service over Swiss Ephemeris, a native library with process-global mutable state that is **not thread-safe** — every call serialised through a single-worker executor that releases the event loop during computation. • Astronomical-timeline subsystem: boundary and instant finders, lunar-month tables, per-year caching with TTL, and typed per-calculation endpoints replacing one overloaded route. **1,078 tests.**
Motherboard API gateway & plugin platform · Go 1.25 · Next.js · Docker	• Architected a Go multi-module monorepo implementing an API gateway and multi-tenant plugin platform — **25 `go.work` modules, 652 tracked Go files**. • Plugin services authenticate, publish routes to a client manager, and the gateway proxies behind a **fail-closed entitlement gate** and a plugin-key claim assertion.
E-commerce Storefront & Embedded Admin Shopify · Liquid · Remix · Prisma	• Developed a Shopify Liquid theme and an embedded Remix admin: metafield/metaobject schema installer, headless catalogue seeders, and certificate verification implemented natively in Liquid rather than through an app proxy. • Accessibility pass to a **44×44** touch-target floor.
Production Portfolio Sites Filmmaker, designer, publisher · Next.js · React 19 · Playwright	• Shipped four production sites. A filmmaker portfolio with a Google Docs→works sync pipeline and multi-codec delivery: a hero codec ladder took a **9.9 MB** H.264 file to **2.3–2.9 MB** across AV1, HEVC and H.264 variants, one downloaded, suppressed entirely for non-watchers, with a **Level 4.0** re-encode because iOS silently refuses to autoplay above 4.2. • A dual-track design portfolio gated by Playwright specs that fail on any hydration warning, plus axe WCAG 2.2 AA and text-spacing checks. A bilingual publishing platform with a separate Express admin carrying a Unicode-script language classifier that replaced an AI one on review. rtistic.vercel.app — Next.js 14, MongoDB, jest and CI, public source.
Git Forensics & Recovery Incident work · git	• Recovered **154 commits** left unreachable in a live repository after a worktree removal combined with dropped stashes — **125 loose commits and 29 stashes** located, tagged under `refs/tags/rescue/*`, verified against `git fsck` and bundled. **8 of 29 stashes** held content present in no branch; one slice was reintegrated behind its tests and the rest deliberately left unrestored, each with a recorded reason. • Separately diagnosed a repository whose `origin` pointed at an **unrelated project with no common ancestor** — 48 commits a single push would have sent to the wrong remote.
Text Digitisation & Translation Pipeline Classical-text corpus · Sanskrit-texts · Python	• Built a **~4,800-line ingestion pipeline** for classical texts — a TEI/SARIT parser detecting division types by containment rather than frequency, a scraper splitting on the source's own headings, five verse-marker grammars detected per source, and per-work structural validators. IAST→Devanagari round-trip verified on **17,086 of 17,090** lines. • One structural fix took depth errors in a single work from **3,522 to 6**. Publishes to a corpus of **92,166 verses across 56 texts**.

Dissertation & Coursework

Dissertation	Human Activity Recognition using an **HMM-based Intermediate Matching Kernel**, representing videos as sequences of sets of feature vectors. Re-implemented in 2026 as a containerised Python pipeline with a Flask front end — github.com/Rupali59/himk-implementation.
Methodology	Classifies a video into a predefined activity class. Feature extraction and classification in Python with scikit-learn, hmmlearn and OpenCV; originally prototyped in MATLAB with Piotr's CV Toolbox, PNetCDF and OpenMPI.
ML Techniques	Bayesian classifiers, Gaussian mixture models, perceptrons, artificial neural networks and support vector machines on linearly separable, non-linearly separable and overlapping data.
Presentations	Presented the components of an information retrieval system and Google's PageRank model at the departmental seminar on Information Retrieval.
Other Projects	Management Information System (MIS), Cybernetics Protector System, Product Management System.

Technical Skills

Languages	Go, TypeScript, JavaScript, Python, PHP, Java, C#, C++, GlidePlatform
Web & Frameworks	Next.js, React, Node.js, Express, Remix, Astro, Laravel, .NET, FastAPI, Tailwind, Bootstrap
AI & Agent Tooling	Claude Code plugins, hooks and skills; structured LLM output and schema-constrained generation; eval harnesses and fixture-based regression suites; hallucination containment and graceful degradation; human-in-the-loop design; MCP
Databases	MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch
Queues & Messaging	RabbitMQ, BullMQ, AWS SQS, Redis Queue
Cloud & DevOps	AWS, Vercel, Fly.io, Docker, Shopify, Jenkins, Git, Linux
ML	scikit-learn, hmmlearn, OpenCV, MATLAB/Octave, pattern recognition and classification

Academic Profile

Year	Degree	Discipline	Institution	Score
2016	**M.Tech**	Computer Science & Engineering Pattern Recognition & Machine Learning	National Institute of Technology, Goa	8.42 CGPA
2014	**B.E.**	Computer Science & Engineering Graduated with Honours	LNCT, Bhopal (RGPV)	8.58 CGPA
2010	XII (CBSE)	Science	St. Joseph's Convent School, Bhopal	83.0%
2008	X (CBSE)	—	St. Joseph's Convent School, Bhopal	88.0%

Certifications

AI — 2026	**Anthropic:** Claude Code in Action; Introduction to Claude Cowork; Claude Platform 101; Claude Code 101; Claude 101. **Google:** Gemini Enterprise Agent Ready; Create Your First Gemini Enterprise Application; Google Cloud Innovator; Developer Program Premium Tier; AI Fundamentals (Google Career Certificates). **Verify:** credly.com/users/rupali-bhatnagar-59 · g.dev/rupali59.
Earlier	WhatsApp API Expert, Wati (2024) · Ultimate JavaScript: Fundamentals, Code With Mosh (2025) · Neural Networks & Deep Learning, Coursera (2020) · JavaScript Algorithms & Data Structures, freeCodeCamp (2020) · Git Essential Training, LinkedIn Learning (2020)

Achievements & Activities

Professional	• Winner, NowFloats Hackathon for product innovation in sales — built the ML-powered SPIDER revenue-prediction module. 2018 • **Geektrust GOLD** — score **95.8** on the Metro Card backend challenge, with **Readability** and **Maintainability** assessed at **ADVANCED** tier by an independent reviewer.
Academic	• **Chancellor's Scholarship (Topper)**, Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal — awarded to the top 10 students statewide by theory marks. 2012, 2013 • **Silver Medal**, Persistent Systems Limited Programming Contest, inter-college. 2013
Other	• Organised the intra-college programming competition, LNCT Bhopal. 2012 • Conducted cultural events, NIT Goa. 2014–2016

Positions of Responsibility

NIT Goa	Teaching Assistant, Computer Science & Engineering Department	2014–2016
NIT Goa	Vice President, Computer Science Department	2014–2016
LNCT, Bhopal	Student Coordinator, Training & Placement Department	2013–2014